Bangladesh University of Business and Technology

LAB REPORT

Experiment No : 05
Course Title : Operating System Lab
Course Code : CSE 210
Experiment Name : Sorting, Prime number, Fibonacci

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Date of Submission : 11/10/2026

Question 1 – Sorting

Steps to solve –

- Open an existing shell script named ‘t.sh’ using ‘vi’
  - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
  - chmod u+x t.sh
- Run the shell script file
  - ./t.sh
- If the value of index[j] is greater than the value of index [j + 1] , then swap (index [j] , index [j + 1])
- Finally the output is shown.

Code –

```
rr=(5 2 7 1 3)
echo  Before sorting: ${rr[@]}
n=${#rr[@]}
for i in $(seq 0 $((n-1)))
do
        for j in $(seq 0 $((n-2)))
        do
                if [ ${rr[j]} -gt ${rr[j+1]} ] ; then
                        temp=${rr[j]}
                        rr[j]=${rr[j+1]}
                        rr[j+1]=$temp
                fi
        done
done
echo After sorting: ${rr[@]}
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Before sorting: 5 2 7 1 3
After sorting: 1 2 3 5 7
```

Question 2 – Prime number

Steps to solve –

- Open an existing shell script named ‘t.sh’ using ‘vi’
  - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
  - chmod u+x t.sh
- Run the shell script file
  - ./t.sh
- The output is shown at the end.

Code –

```
read -p "Enter a number: " a
if [ $a -le 1 ] ; then
        echo $a is not prime
        exit 0
fi
for i in $(seq 2 $((a-1)))
do
        if [ $((a%i)) -eq 0 ] ; then
                echo $a is not prime
                exit 0
        fi
done
echo $a is prime
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number: 10
10 is not prime
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number: 11
11 is prime
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number: 13
13 is prime
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number: 17
17 is prime
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number: 9
9 is not prime
```

Question 3 – Fibonacci

Steps to solve –

- Open an existing shell script named ‘t.sh’ using ‘vi’
  - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
  - chmod u+x t.sh
- Run the shell script file
  - ./t.sh
- The output is shown at the end.

Code –

```
read -p "Enter input: " a b
read -p "Enter number of terms: " terms
for i in $(seq 1 $terms)
do
        echo $a
        c=$((a+b))
        a=$b
        b=$c
done
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter input: 0 1
Enter number of terms: 10
0
1
1
2
3
5
8
13
21
34
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$
```

Question 4 – $1^2+3^2+5^2+7^2$

Steps to solve –

- Open an existing shell script named ‘t.sh’ using ‘vi’
  - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
  - chmod u+x t.sh
- Run the shell script file
  - ./t.sh
- Here number 7 has been taken as input ‘terms’
- The output is calculated through the equation : sum= sum + i^2
  - At first the value of sum is equal to 0
  - At iteration,
    i. sum = 0 + 1^2 = 1 ;
    ii. sum = 1 + 3^2 = 1 + 9 =10 ;
    iii. sum = 10 + 5^2 = 10 + 25 = 35 ;
    iv. sum = 35 + 7^2 = 35 + 49 = 84 ; (and the loop ends)
- At last the output 84 is given.

Code –

```
read -p "Enter number of terms: " terms
sum=0
for i in $(seq 1 2 $terms )
do
        sum=$((sum+$((i*i))))
done
echo $sum
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter number of terms: 7
84
```